

Hepatic Proteome and Its Network Response to Supplementation of an Anti-obesity Herbal Mixture in Diet-induced Obese Mice

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Abstract In the present study, we investigated the effects of multi-herbal water extract mixture, Taeumjowi-tang (TH) on liver proteome alteration in mice using two-dimensional electrophoresis combined with MALDI-TOF-MS. Animals were fed high-fat diet with or without TH (0.3% wt/wt) supplement for 12 weeks. At the end of 5th week of experimental diet, mice fed high-fat diet only were subdivided into 2 groups, obesity-prone (OP) and obesity-resistant (OR) mice based on weight gain. OR mice gained less body weight compared to OP mice despite of same food intake. TH significantly suppressed weight gain, and proteomic analysis enabled the identification of 49 liver proteins showing differential regulation between OP and OR/TH mice. Combined results of proteomic and western blot analyses revealed decreased lipogenesis *via* three fatty acid metabolic targets (AMPK, ACC, and FAS) in livers of OR and TH mice. Using bioinformatic classification and network analysis, most of the identified proteins were classified as hydrolases, oxidoreductases, transferases, defense/immunity proteins, and enzyme modulators based on functional analysis of the PANTHER classification system. Combined results of proteomic and bioinformatic

analyses using GeneMANIA identified two proteins (LACTB2 and NIT2) in the liver that potentially interact with fatty acid metabolic proteins. Furthermore, these proteins were included in acetylation, phosphoprotein, and metabolic processes in DAVID classification. These proteins were highly expressed in OP mice; however both their transcription and protein expression were lowered by TH treatment. In conclusion, combined data from proteomic and network analyses suggest that TH exerts anti-obesity effects by modulating fatty acid metabolic proteins/genes, particularly *via* the AMPK pathway. Most targeted proteins/genes were modulated toward enhancing lipid metabolism in response to TH treatment.

Keywords: anti-obesity, obesity susceptibility, obesity resistance, proteome, liver

1. Introduction

Excessive energy intake is associated with increasing rates of obesity as well as non-alcoholic fatty liver disease [1,2]. In particular, accumulation of excess intra-hepatic triglycerides from consumption of a high fat diet (HFD) is a cause of metabolic and nutritional disorders [3]. Increasing evidence indicates that an HFD leads to progression of non-alcoholic fatty liver disease in rodents and humans [4–8]. Among many obese animal models, HFD-induced C57BL/6J mice have been extensively used [9,10]. However, most animal models display large inter-individual differences in obesity development upon prolonged exposure to an HFD. For instance, obesity-prone (OP) or obesity-resistant (OR) subjects differ markedly in their regulation of food intake and metabolic efficiency upon consumption of an HFD

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[11]. Previously, we identified several important proteins from the blood and metabolic tissues of rats associated with phenotypic differences relating to obesity susceptibility and resistance in a series of proteomic studies [12-14]. However, the underlying mechanism for these phenotypic differences remains to be understood [15], especially with regard to anti-obesity herbal mixtures and the synergistic effects of multi-compounds included in those herbal mixtures.

The focus of this current obesity research is the prevention and management of diet-induced obesity in response to a multi-herbal mixture. Single target-based therapy has limitations in obesity management, since high calorie intake only affects a few key proteins/genes and functionally important biomolecules [16]. For this reason, medicinal herbal extracts can be a promising alternative strategy for the treatment of obesity due to their lower toxicity and multi-targeting capacity with synergistic effects against biomolecules involved in obesity development [17,18]. Reports on anti-obesity herbal extract formulations greatly outnumber preclinical and clinical studies conducted on obesity and non-alcoholic fatty liver disease [19-21]. The treatment of natural products for metabolic diseases has not been explored in depth despite the fact that oral hypoglycemic agents, such as metformin, are derivatives of natural plant products [22,23]. Recently, the application of natural medicines or medicinal food for the treatment of liver disease and metabolic syndromes has been increasing in Asian countries because of their ability to protect hepatocytes through inhibiting hepatic inflammation and fibrosis in the liver [24-26]. Although several traditional medicines have been reported to be useful for the treatment of liver diseases, the molecular targets of such compounds have scarcely been identified in animal models [25,26]. Among numerous anti-obesity herbal medicines, Taeumjowitang (TH) is a decoction consisting of eight herbal ingredients which has traditionally been used as an herbal medicine in Korea to treat dyspepsia, stuffiness, the sensation of fullness, diarrhea, and wind stroke [27-30]. Earlier reports have demonstrated effects of TH and its modified prescription on obesity and hyperlipidemia [27-32] but mechanistic studies are needed for its therapeutic application.

Proteomics technology allowing for proteome profiling between lean and obese subjects has been well accepted for the identification of biomarkers associated with obesity [12,33-37]. In particular, two-dimensional electrophoresis (2-DE) combined with MALDI-TOF-MS provides a strong technological tool for the separation and identification of target proteins for obesity treatment [38,39]. Several proteomic studies in the liver have been conducted to identify responsive marker proteins to various nutritional conditions [9,12,40].

In the present study, we investigated the effects of TH,

an anti-obesity herbal medicine, on liver proteome alteration in mice fed an HFD using 2-DE electrophoresis combined with MALDI-TOF-MS. In addition, differential hepatic proteomic data were compared between OP and OR mice. Further, network analyses of liver proteome in response to anti-obesity herbal medicine (TH) were applied. The current study may be a useful to evaluate the effects of natural compounds on multi-target based therapy for obesity and other metabolic disorders.

2. Materials and Methods

2.1. Animals and breeding conditions

Four-week-old C57BL/6J male mice were purchased from Jackson Laboratory (Bar Harbor, ME, USA). All mice were individually housed under constant temperature (24°C) and a 12 h light/12 h dark cycle. Mice were provided water and standard chow *ad libitum* for 1 week for acclimatization before the start of experimental procedures. At 5 weeks of age, mice were randomly divided into three groups, normal diet control group (ND, $n = 10$), HFD-fed control group (HFD, 60% fat, $n = 20$), and TH-supplemented HFD group (TH, 60% fat, $n = 10$). After 6 weeks on the HFD, 20 mice were further subdivided into 2 groups based on weight gain profile: HFD-OP group ($n = 10$) and HFD-OR group ($n = 10$). Mice were fed the experimental diets for entire 12 weeks. The herbal medicinal formula, TH, was purchased from I-world Pharm. Co. Ltd. (Incheon, Korea) in a powdered form of its water extract. TH decoction is formulated with eight herbal ingredients: *Semen coicis* (11.25 g), *Semen castaneae* (11.25 g), *Semen raphani* (7.5 g), *Schisandrae fructus* (3.75 g), *Liriopsis tuber* (3.75 g), *Herba ephedrae*

Table 1. Compositions of normal diet (ND), high-fat diet (HFD), and high-fat diet with Taeumjowitang (TH)

Composition by weight, g/kg	ND	HFD	TH
Casein	200	265	265
Corn starch	397.486	0	0
Sucrose	100	90	60
Maltodextrin	132	160	160
Cellulose	50	65.6	65.6
Soybean oil	70	30	30
Lard	0	310	310
Mineral mix	35	48	48
Calcium phosphate, dibasic	0	3.4	3.4
Vitamin mix	10	21	21
TBHQ antioxidant	0.014	0	0
L-cystine	3	4	4
Choline bitartrate	2.5	3	3
Taeumjowitang			30
Total (kcal/kg)	3,800	5,100	4,980

(3.75 g), *Radix platycodi* (3.75 g), and *Acori tatarinowii rhizoma* (3.75 g). Compositions of the experimental diets are presented in Table 1. Mice had free access to food and distilled water during the experimental period. Food intake and body weights were measured daily and weekly, respectively. On the 12th week, mice were anaesthetized with diethyl ether and sacrificed after 12 h of fasting. Blood was taken from the *inferior vena cava* and centrifuged at $1,000 \times g$ for 15 min at 4°C, after which plasma was isolated to characterize plasma biochemistry. After blood collection, epididymal adipose tissues were promptly removed, rinsed, weighed, frozen in liquid nitrogen, and stored at -80°C. The animal study was approved by the Ethics Committee for Animal Studies at Kyungpook National University, Republic of Korea.

2.2. Preparation of protein samples

Whole liver was excised from mice immediately following anesthetization after overnight fasting. Liver tissues were then washed with cold saline solution and pulverized under liquid nitrogen and stored at -80°C. Tissues were lysed as previously described [12]. The supernatant was then stored at -80°C until analysis. Protein content of whole liver tissue was determined using the RC DCTM protein assay (Bio-Rad).

2.3. Two-dimensional electrophoresis (2-DE)

2-DE was performed in duplicate using liver protein samples from six mice per diet group: ND, OP, OR, and TH mice. 2-DE experiments were conducted using the previous methods published by our laboratory [36,41,42]. Briefly, immobilized pH gradient (IPG) -isoelectric focusing (IEF) of liver samples was performed using pH 4-7 and 18-cm IPG DryStrips (GE Healthcare, Buckinghamshire, UK) in a PROTEIN IEF cell (Bio-Rad) following manufacturer's protocol. IPG strips were passively rehydrated for 12 h in strip holders with 350 μ L of rehydration solution (7 M urea (Bio Basic, Ontario, Canada), 2 M thiourea (Sigma, St. Louis, MO, USA), 4% CHAPS (Bio Basic), 1 mM PMSF (Sigma), 20 mM DTT (GE Healthcare), 2% IPG buffer (Bio-Rad), and 150 μ g of liver protein). IEF was executed as previously described [12]. Gel strips were then placed onto a 20 $\times$ 20 cm 12% polyacrylamide gel for resolution in the second dimension. Fractionation was performed using a Laemmli SDS discontinuous system at a constant voltage of 15 mA per gel for 14 h. For image analysis and peptide mass fingerprinting (PMF), a total of 48 gels (two gels per diet group with separated proteins) were visualized by silver staining [12].

2.4. Image acquisition and data analysis

Gels were imaged on a UMAX PowerLook 1120 (Maxium

Technologies, Akron, OH, USA), and modified ImageMaster 2-D software V4.95 (GE Healthcare) was used for comparison of images. A reference gel was selected from the gels of the normal group, and the detected spots from other gels were matched with those of the reference gel. Relative optical density and relative volume were calculated in order to correct differences within gel staining. Each spot intensity volume was calculated by background subtraction and total spot volume normalization. The resulting spot volume percentage was used for comparison.

2.5. Protein identification

For protein identification by PMF, protein spots were excised, digested with trypsin (Promega, Madison, WI, USA), mixed with CHCA in 50% ACN/0.1% TFA, and subjected to MALDI-TOF analysis (Microflex LRF 20, Bruker Daltonics). Spectra were collected from 300 shots per spectrum over an m/z range of 600 ~ 3,000 and calibrated by two-point internal calibration using trypsin auto-digestion peaks (m/z 842.5099, 2211.1046). The peak list was generated using Flex Analysis (ver 3.0). The threshold used for selecting peaks was as follows: 500 for minimum resolution of monoisotopic mass, 5.0 for S/N. Peptide masses were matched with the theoretical peptides of all proteins in the NCBI database using MASCOT (Matrixscience, <http://www.matrixscience.com>). The following parameters were used for the database search: trypsin as the cleaving enzyme, maximum of one missed cleavage, iodoacetamide (Cys) as complete modification, oxidation (Met) as partial modification, monoisotopic masses, and mass tolerance of ± 0.1 Da. Protein score is $-10 \times \log(p)$, where p is the probability that the observed match is a random event, and greater than 61 is significant ($p < 0.05$).

2.6. Immunoblot analysis

Immunoblot analyses were performed as previously described [12]. Briefly, tissue lysates were prepared with RIPA buffer (Sigma), homogenized, and centrifuged at $12,000 \times g$ for 20 min. Extract was diluted in sample buffer [12]. Membranes were subsequently blocked with 5% non-fat dry milk in TBS (10 mM Tris-HCl, 150 mM NaCl, pH 7.5) containing 0.1% Tween-20 (TBS-T). After washing with TBS-T, membranes were incubated for 2 h with a 1:1,000 dilution of primary polyclonal mice antibody (anti-ACC, anti-AMPK, anti-FAS, anti-LACTB2, anti-NIT2, anti-pACC, anti-pAMPK, or anti- β -actin; Santa Cruz Biotechnology, Santa Cruz, CA, USA), HRP-conjugated anti-rabbit IgG, anti-goat IgG, or anti-mouse IgG secondary antibody (1:1,000; Santa Cruz Biotechnology), and developed using enhanced chemiluminescence (iNtRON Biotechnology, Seoul, Korea).

2.7. Network analyses

For gene/protein classification and network analysis, identified proteins obtained from proteomic analysis together with major fatty acid metabolic proteins were analyzed using Protein ANnotation THrough Evolutionary Relationship (PANTHER) [43] and Database for Annotation, Visualization, and Integrated Discovery (DAVID) [44] classification systems. These public databases make it possible to systematically divide large gene lists in an attempt to assemble a summary of the most related biological functions, pathways, and protein classes. Identification of gene enrichment categories was determined according to Swiss-Prot (SP) - Protein Information Resource (PIR) keywords that combined the annotation from both SP and PIR. Finally, associations of differentially expressed genes with broadly defined molecular networks were carried out using the GeneMANIA (<http://www.genemania.org>) web interface [45]. This interface generates a list of genes with functional similarity based on the available genomics and proteomics database. There are two important parts of the GeneMANIA algorithm: a linear regression-based algorithm for calculating a single, composite functional association network from multiple networks derived from different proteomic or genomic data sources; and prediction of gene function.

2.8. Real-time RT-PCR

Transcription levels of genes were quantitatively determined by real-time RT-PCR (Stratagene Mx 3000p QPCR System, Santa Clara, CA, USA). Total RNA was isolated from liver tissues using an easy-spinTM (DNA-free) Total RNA Extraction kit (Company, City, Country), and reverse-transcription was carried out using a Maxime RT Premix kit (iNtRON Biotechnology). FastStart Universal SYBR Green Master (Roche Diagnostics, Indianapolis, IN,

USA) was employed for real-time RT-PCR. Transcription levels of each gene (*Acaca*, *Fasn*, *Lactb2*, *Nit2*, and *Prkaa1*) were normalized to *Gusb* transcription levels. Oligonucleotide sequences are shown in Table 2.

2.9. Statistical analysis

All experimental results were compared by One-way Analysis of Variance (ANOVA) using the Statistical Package of Social Science (SPSS, version 14.0K) program; data are expressed as the mean $\pm$ SEM. A protected least-significant difference (LSD) test, a method for multiple comparisons consisting of single-step procedures in One-way ANOVA, was used to demonstrate significant differences between means.

3. Results

3.1. Phenotypic differences in mice fed an HFD and a TH-supplemented HFD

Changes in body weight among the four mice groups after 12 weeks are shown in Fig. 1. Body weights of mice were mostly the same among all groups at the beginning of the study (data not shown). However, final body weight gain of HFD mice (OP) was significantly higher (33%) compared to OR mice (Fig. 1A). No significant difference in food intake was observed among HFD-fed groups (OP, OR, and TH), suggesting that body weight reduction in OR and TH groups was not due to less caloric intake (Fig. 1B). Additionally, total white adipose tissue (WAT) weight per body weight after TH treatment remarkably decreased by up to 15% ($p < 0.01$) compared to OP mice (Fig. 1C). Consequently, these data show that TH inhibits fat accumulation in HFD-fed mice. Further, circulating concentrations of total cholesterol and hepatic triglycerides were significantly higher in OP mice compared to ND, OR, and TH mice ($p < 0.05$) (Figs. 1D and 1E). However, no significant difference was observed in plasma triglyceride levels between OP and OR mice (Fig. 1D). Levels of circulating glucose, glucagon, and insulin were significantly higher in OP mice compared to OR and TH groups ($p < 0.05$) (Figs. 1F, 1G and 1H).

3.2. Differential expression of liver proteins in response to HFD and TH treatment

Total liver proteins were separated by 2-DE using a pH 4-7 IEF strip with 12% (w/v) SDS-PAGE gels. More than 1,000 matched spots were obtained in all groups (Fig. 2), and 49 spots were identified by MALDI-TOF-MS (Table 3). All spots were identified with high confidence based on high scores and sequence coverage. Among the 49 differentially expressed spots, 12 spots were significantly

Table 2. List of RT-PCR primers

Genes		Primer sequence (5'-3')
<i>Acaca</i>	F ^a	TGGCGTCCGCTCTGTGATAG
	R ^b	ACAACCTCTGCTCGCTGGGT
<i>Fasn</i>	F	CTGATGTGACAACGAGCCGC
	R	GACAGGCACTCAGCCTCCAT
<i>Lactb2</i>	F	GGCAGAGCATAATCTCTTGCTTCA
	R	CACTGATCTGTAATAAGGACGGC
<i>Nit2</i>	F	TCGCGGAGCTTGACAAATC
	R	GTCATCCCGAGCAGGAGAGG
<i>Prkaa1</i>	F	CCATGCGCAGACTCAGTTCC
	R	ATGTCCGGTCAACTCGTGCT
Internal control		
<i>Gusb</i>	F	CGGGACTTTATTGGCTGGGT
	R	CCATTCACCCACACAACTGC

^aF: sequence from sense strands; ^bR: sequence from anti-sense strands.

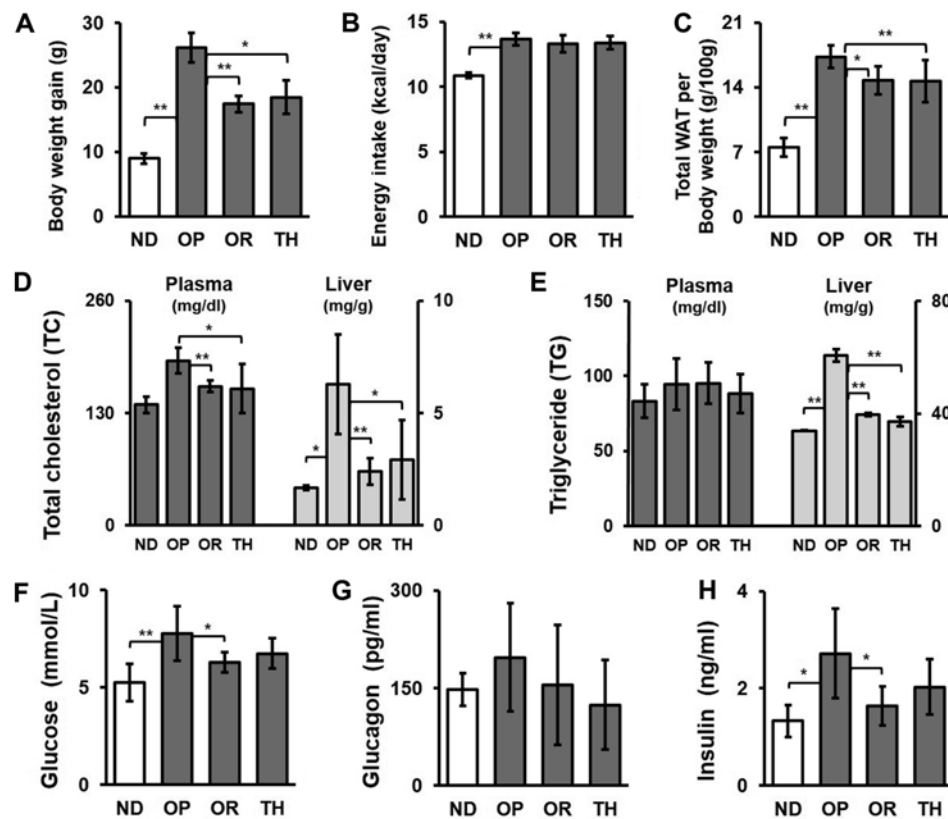


Fig. 1. (A) Average body weight gain, (B) energy intake, (C) average weight of white adipose tissue (WAT) per body weight, (D) total cholesterol, (E) triglyceride, (F) glucose, (G) glucagon, and (H) insulin in mice fed normal diet (ND), HFD-fed obesity-prone mice (OP), HFD-fed obesity-resistant mice (OR), and HFD-fed obese mice treated with the herbal medicine Taemjowi-tang (TH). Statistical significance was determined by One-way ANOVA, where p values are * p < 0.05 and ** p < 0.01 for significance. Error bars represent means $\pm$ S.E.M (n = 6).

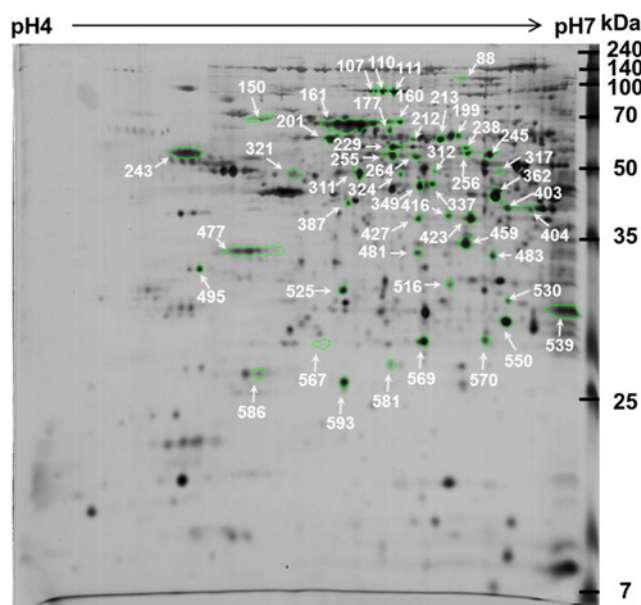


Fig. 2. Representative silver-stained 2-DE gel image of mice liver proteome. Differentially regulated proteins and proteins of interest are marked with circles and arrows. Numbers in gels with their corresponding protein names are listed in Table 3.

down-regulated in OP mice when compared to ND mice (Fig. 3A), whereas 37 spots showed significantly higher levels in OP mice (Fig. 3B).

3.3. Network analysis and classification of liver proteins

Most of the identified proteins were related to metabolic and cellular processes in the PANTHER Classification Database (Fig. 4A). According to the results of protein classification, most identified proteins were classified as hydrolases, oxidoreductases, transferases, defense/immunity proteins, and enzyme modulators (Fig. 4B). Functional analysis of identified proteins was also performed using DAVID Bioinformatics Resources (Fig. 4C). The results are uniquely categorized into five major protein classes (phosphoprotein, acetylation, cytoplasm, metal-binding, and hydrolase), according to their underlying enrichment algorithms. To visualize gene interactions, we performed network analysis using GeneMANIA to determine *in silico* gene interactions for the identified proteins by proteomic analysis. Interactions among query genes were constructed, and a close-up screenshot was captured with all other interactions hidden (Figs. 5A and 5B).

Table 3. Differentially expressed hepatic proteins (2≥fold change) in normal diet (ND)-fed, high fat diet (HFD)-treated obesity-prone (OP), HFD-fed obesity-resistant (OR), and HFD-fed obese mice treated with Taeumjowi-tang (TH)

Spot ID	Description	Acc. No. ^a	Nominal mass (M _r) ^b	Calculated PI	Score ^c
88	Pyruvate carboxylase, mitochondrial isoform 2	gi 251823978	130360	6.25	227
107	Cytosolic 10-formyltetrahydrofolate dehydrogenase	gi 27532959	99502	5.64	368
110	Cytosolic 10-formyltetrahydrofolate dehydrogenase	gi 27532959	99502	5.64	368
111	Cytosolic 10-formyltetrahydrofolate dehydrogenase	gi 27532959	99502	5.64	368
150	Pdia4 protein, partial	gi 45219865	65388	5.91	139
160	Serum albumin precursor	gi 163310765	70700	5.75	106
161	Serum albumin precursor	gi 163310765	70700	5.75	155
177	Myosin-1	gi 82524274	224116	5.60	71
199	Carboxylesterase 3B isoform 2 precursor	gi 226874916	57810	5.84	67
201	Hspd1 protein, partial	gi 76779273	59388	8.09	212
212	Epoxide Hydrolase	gi 563510	62505	5.85	102
213	Epoxide Hydrolase	gi 563510	62505	5.85	107
229	Formimidoyltransferase-cyclodeaminase	gi 18252784	59529	5.79	141
238	T-complex protein 1 subunit beta	gi 126521835	57441	5.97	150
243	Protein disulfide-isomerase precursor	gi 42415475	57422	4.77	231
245	Alpha-aminoadipic semialdehyde dehydrogenase isoform b	gi 188035915	55610	5.99	129
255	Selenium-binding protein 2	gi 9507079	52594	5.78	173
256	Alpha-aminoadipic semialdehyde dehydrogenase isoform b	gi 188035915	55610	5.99	136
264	Selenium-binding protein 1	gi 22164798	52480	5.87	183
311	S-adenosylmethionine synthase isoform type-1	gi 19526790	44051	5.51	190
312	Phenylalanine hydroxylase	gi 148689509	50265	5.91	94
317	Eno1 protein	gi 34784434	39757	5.86	82
321	Endo B cytokeratin	gi 293685	47475	5.22	134
324	Proteasome (prosome, macropain) 26S subunit, ATPase 2, isoform CRA_a	gi 148671241	44694	5.83	75
337	Doublecortin domain-containing protein 2	DCDC2_MOUSE	51978	6.11	50
349	Ornithine aminotransferase, mitochondrial precursor	gi 8393866	48723	6.19	263
362	S-adenosyl-L-homocysteine hydrolase	gi 904132	47643	5.98	126
387	ASL1/Suc1g2 fusion protein	gi 188038847	45039	5.75	217
403	Sorbitol dehydrogenase	gi 22128627	38225	6.56	150
404	Sorbitol dehydrogenase	gi 22128627	38225	6.56	112
416	Fructose-1,6-bisphosphatase 1	gi 9506589	36889	6.15	115
423	Fructose-1,6-bisphosphatase 1	gi 9506589	36889	6.15	141
427	Fructose-1,6-bisphosphatase 1	gi 9506589	36889	6.15	120
459	Malate dehydrogenase, cytoplasmic	gi 254540027	36488	6.16	90
477	Regucalcin	gi 6677739	33385	5.15	147
481	Ester hydrolase C11orf54 homolog	gi 19526926	35430	5.86	54
483	Quinolinate phosphoribosyltransferase, isoform CRA_b	gi 148685554	31509	6.71	78
495	Anxa5 protein	gi 13277612	35716	4.83	163
516	Beta-lactamase-like protein 2	gi 21703764	32734	5.89	98
525	Prohibitin	gi 6679299	29802	5.57	220
530	Omega-amidase NIT2	gi 12963555	30483	6.44	106
539	Carbonic anhydrase 3	gi 31982861	29348	6.89	114
550	Peroxiredoxin-6	gi 6671549	24811	5.98	198
567	Dual specificity protein phosphatase 13 isoform B	DS13B_MOUSE	22581	9.79	47
569	Alpha/beta hydrolase domain-containing protein 14B	gi 171460960	22437	6.04	120
570	Casein kinase II subunit alpha	CSK21_MOUSE	45219	7.29	41
581	Sterile alpha motif domain-containing protein 14	SAM14_MOUSE	45303	9.51	44
586	Lactoylglycyl-L-histidyl-L-histidine lyase	gi 165932331	20967	5.24	70
593	ATP synthase subunit d, mitochondrial	gi 21313679	18795	5.52	142

^aNCBI/TrEMBL database accession number. ^bThe nominal mass is the integer mass of the most abundant, naturally occurring stable isotope of an element. ^cMASCOT probability-based molecular-weight search score calculated for peptide mass fingerprinting. Protein score is $-10 \times \log(P)$, where P is the probability that the observed match is a random event; it is based on the NCBI/TrEMBL database using the MASCOT searching program as MS/MS data and protein scores >61 are significant ($P < 0.05$). ^dND means not detected. Spot Id is same numbers in gel image of Fig. 2.

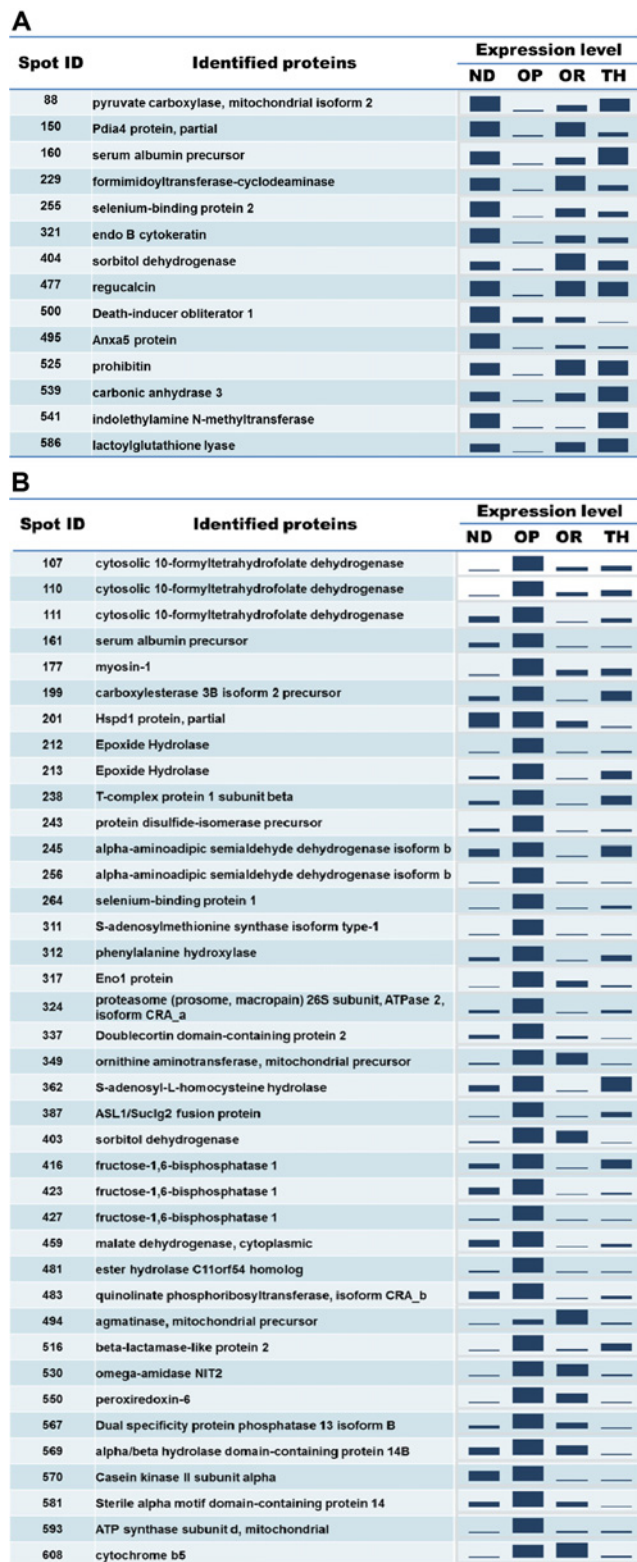


Fig. 3. Differentially regulated proteins in liver of OP mice compared with ND/OR/TH mice. (A) Down-regulated and (B) up-regulated proteins in OP mice. Expression levels depicted as bars of each protein are presented as their average volume density (%) in 2-DE analysis ($p < 0.05$).

3.4. Interactions between identified proteins with fatty acid metabolism

To elucidate a correlation between the identified proteins and fatty acid metabolism in the liver, we investigated three metabolically important proteins (ACC, AMPK, and FAS) in the liver that were not detected in the proteomic analysis. To this end, we focused on interacting partners of these fatty acid metabolic proteins with identified proteins using GeneMANIA network analysis (Figs. 6, 7 and 8). When candidate query genes from identified proteins were subjected to analysis, eight genes were predicted as related to *Prkaa1* (AMPK) and *Acaca* (ACC) (Fig. 6). Among the eight partners of *Prkaa1*, *Hspd1* and *P4hb* were categorized as chaperones; *Anxa5*, *Csnk2a1*, *Myh1*, *Nit2*, *Hspd1*, and *Aldh1l1* were categorized as phosphoproteins; *Csnk2a1*, *Myh1*, and *Hspd1* were categorized as ATP-binding proteins in SP_PIR_KEYWORD from DAVID webserver (Fig. 6). Moreover, eight targets with two major fatty acid metabolic genes (*Fasn* and *Prkaa1*) were closely connected with *Acaca* (Fig. 7). These genes were also shown to be involved in acetylation (*Csnk2a1*, *Glo1*, *Krt18*, *Atp5h*, and *Pcx*), mitochondrial function (*Oat*, *Atp5h*, and *Pcx*), and lyase (*Car3* and *Glo1*) in DAVID functional categories. We also detected 12 genes as closely correlated with *Fasn* (FAS) and *Acaca* (Fig. 8). Among these 12 genes, most genes were shown to be involved in acetylation (*Glo1*, *Lactb2*, *Mdh1*, *Hspd1*, *Phb*, *Pcx*, and *Sord*), oxidoreduction (*Mdh1*, *Aldh1l1*, and *Sord*), metal-binding (*Car3*, *Glo1*, *Lactb2*, *Pcx*, and *Sord*), and mitochondrial function (*Hspd1*, *Phb*, and *Pcx*) among DAVID bioinformatics resources. Six genes were shown to be associated with chaperone function (*Hspd1* and *P4hb*), lyase activity (*Car3* and *Glo1*), and NAD metabolism (*Mdh1* and *Sord*) in DAVID. Combined results of proteomics with network analysis revealed that most identified genes are correlated with fatty acid metabolic genes via the AMPK pathway, and TH plays a preventive role in hepatic steatosis of obesity.

3.5. Validation of network analysis

To evaluate the results of network analysis, expression levels of two potential marker genes/proteins identified in this study (*Lactb2*/LACTB2 and *Nit2*/NIT2), together with key fatty acid metabolic genes *Acaca*, *Fasn*, and *Prkaa1*, were further confirmed using real-time RT-PCR and Western blot analysis, respectively (Fig. 9). Expression levels of the two potential targets (*Lactb2*/LACTB2 and *Nit2*/NIT2) dramatically increased in response to HFD treatment along with changes in fatty acid metabolic genes/proteins, whereas their expression significantly decreased in the livers of OR and TH-treated mice. Expression patterns of all these proteins are in line with the proteomic results.

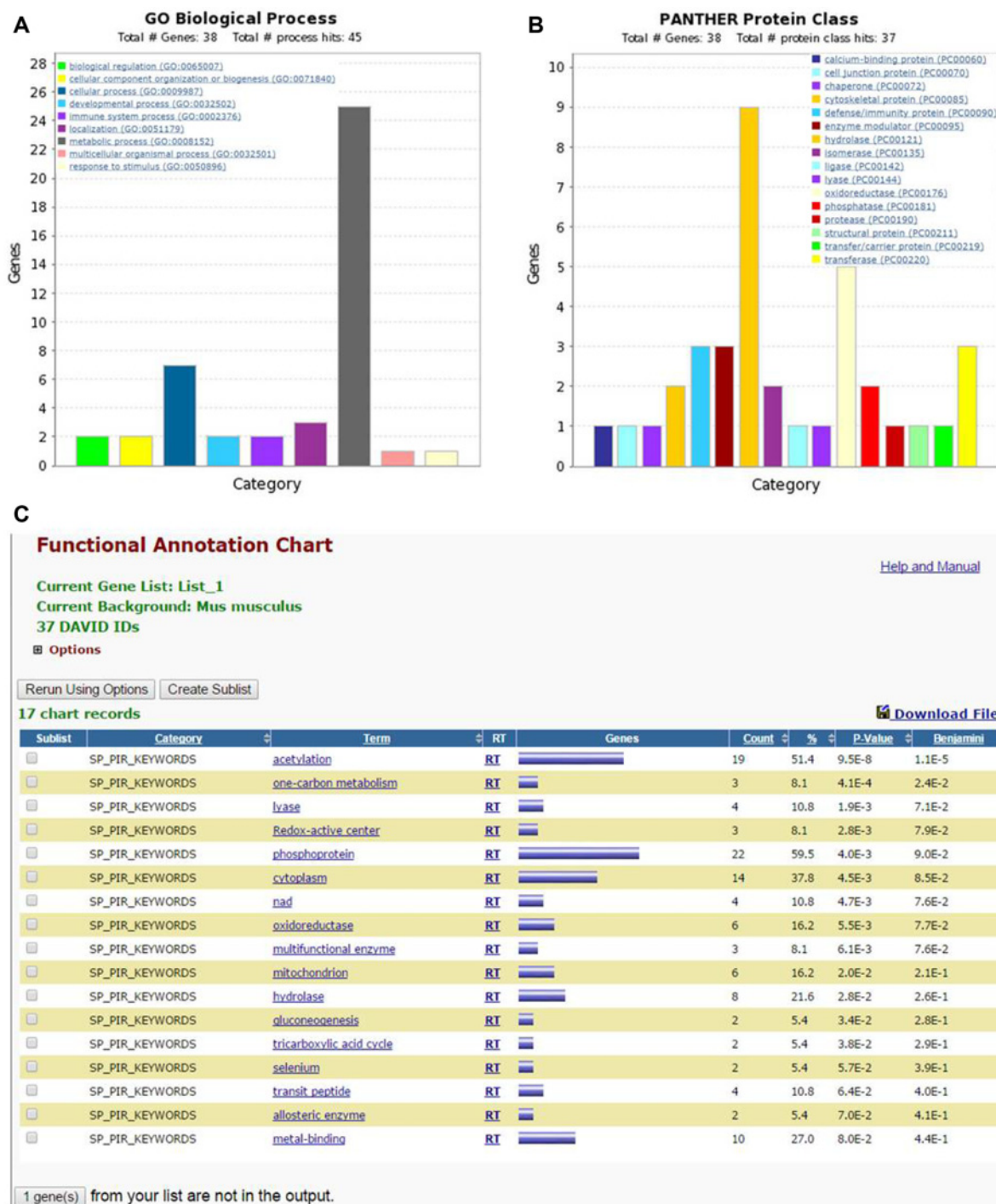


Fig. 4. Classifications of biological process and protein class using PANTHER classification system and DAVID bioinformatics resources: (A) biological process, (B) protein class, and (C) functional annotation chart. **Chart tooltips are read as: Category name (Accession): # genes; Percent of gene hit against total # genes; Percent of gene hit against total # Process/Protein class hit.

4. Discussion

In the present study, we investigated the effects of an anti-obesity herbal medicine (TH) on liver proteome alteration in obesity-prone (OP) and obesity-resistant (OR) mice fed

an HFD and identified potential target proteins associated with obesity. In addition, bioinformatics classification and network analysis were carried out to investigate functional connections between identified proteins and key fatty acid metabolic targets (AMPK, ACC, and FAS).

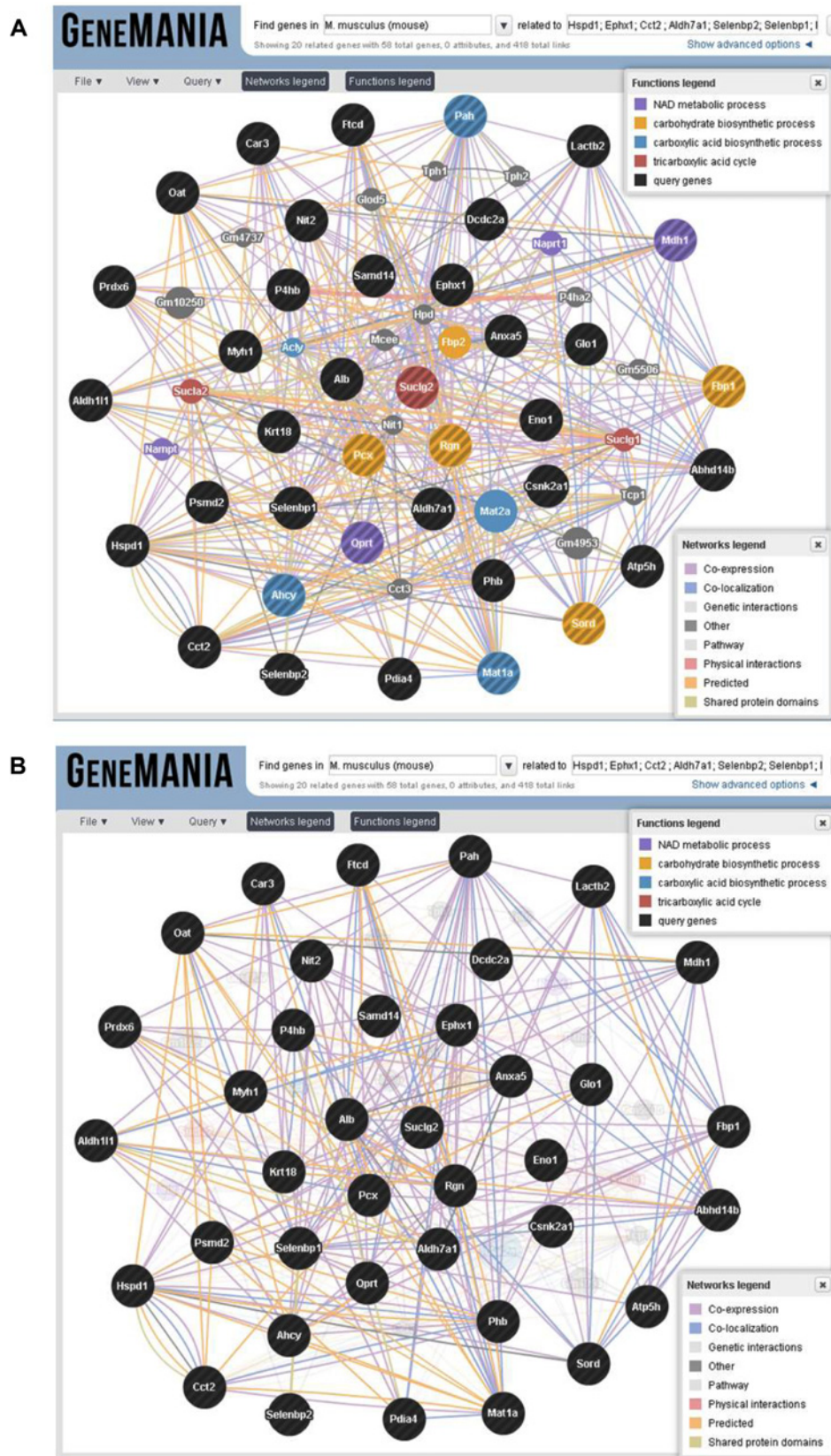


Fig. 5. GeneMANIA showing results of 38 queries (encircled with different colors and bolded) from livers of mice using the advanced parameter: molecular function-based. (A) Extreme right panel depicts functions of the given queries, (B) GeneMANIA result window depicts close-up of interaction of only query genes.

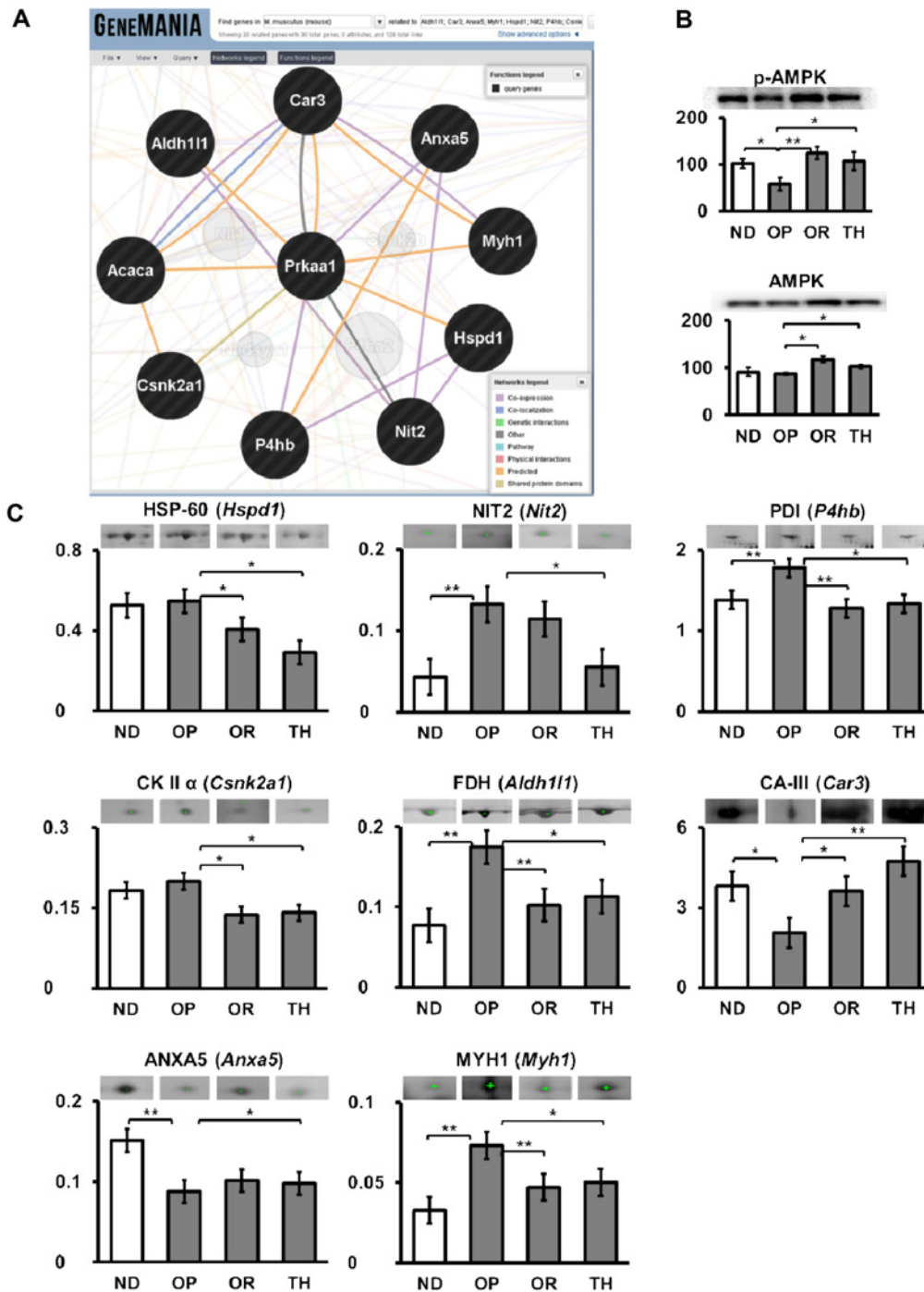


Fig. 6. Identification of differentially expressed liver proteins linked to *Prkaa1* (AMPK) expression. (A) GeneMANIA result window depicts a close-up of the interaction with *Prkaa1* (AMPK), (B) Western blot analysis of AMPK/p-AMPK, (C) differential expression of identified proteins from proteomic analysis. Band density of Western blot image was calculated using KODAK 1D Image Analysis software. Relative intensity (%) demonstrated that protein levels were normalized to that of β -actin. Data shown are representative of three independent experiments. Statistical significance was determined by One-way ANOVA, where p values are $*p < 0.05$ and $**p < 0.01$ for significance.

It has been demonstrated that HFD intake leads to diet-induced obesity along with many disease risk factors in association with increased post-translational modification

(PTM), production of advanced glycation end products, and apoptosis [46,47]. Our proteomic data confirmed five proteins having the same molecular weight but different

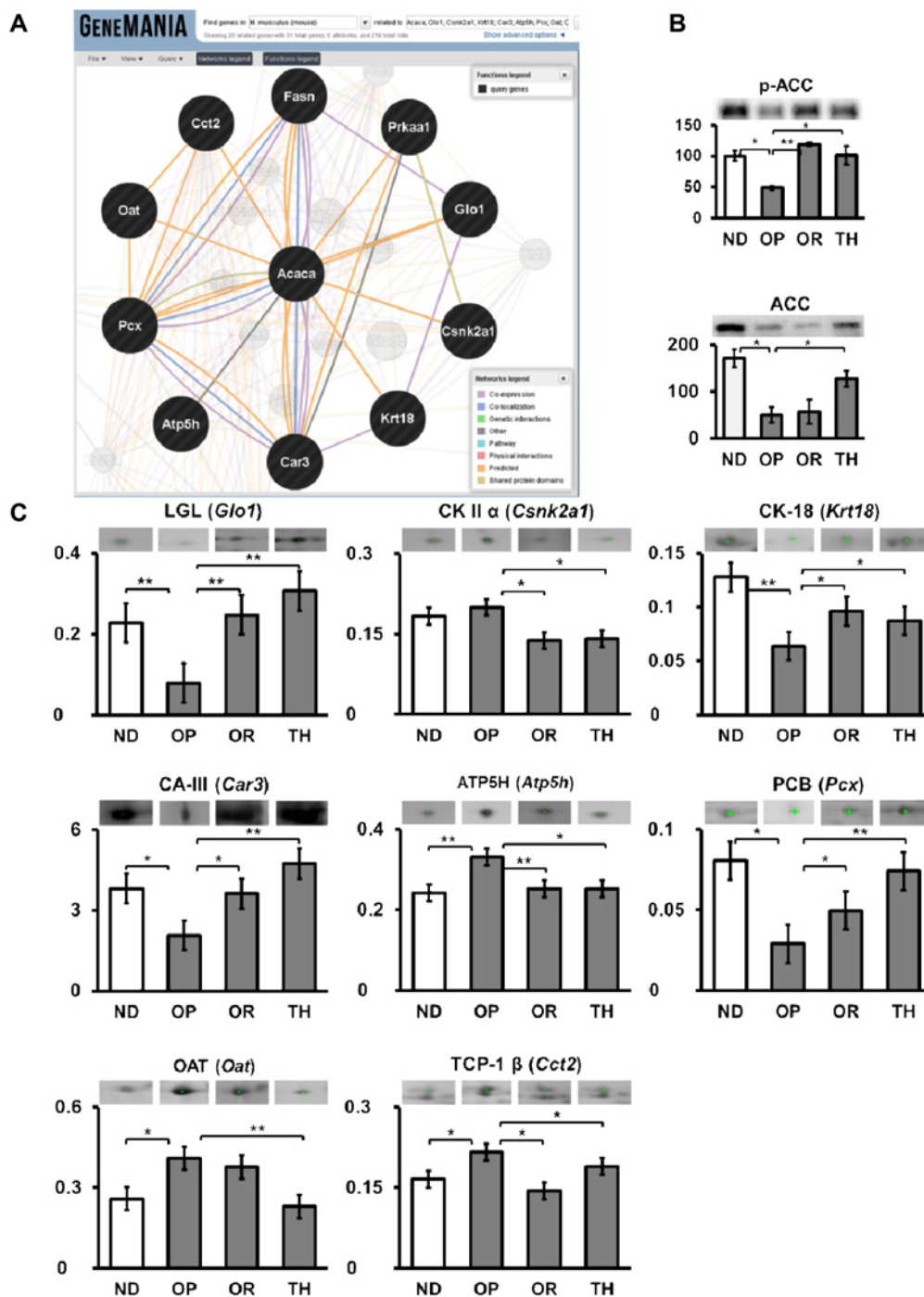


Fig. 7. Identification of differentially expressed liver proteins linked to *Acaca* (ACC) expression. (A) GeneMANIA result window depicts close-up of interaction with *Acaca* (ACC), (B) Western blot analysis of ACC/p-ACC, (C) differential expression of identified proteins from 2-DE. Band density of Western blot image was calculated using KODAK 1D Image Analysis software, and relative intensity (%) demonstrated that protein levels were normalized to that of β -actin. Data shown are representative of three independent experiments. Statistical significance was determined by One-way ANOVA, where p values are * p < 0.05 and ** p < 0.01 for significance.

pl, indicating the proteins underwent PTM. Among those five proteins, alpha-aminoadipic semialdehyde dehydrogenase (α -AASA dehydrogenase) and cytosolic 10-formyltetrahydrofolate dehydrogenase (FDH) showed higher expression

levels in HFD-induced obese mice. α -AASA dehydrogenase is a multifunctional enzyme that mediates important protective effects in cells against oxidative stress by metabolizing lipid peroxidation-derived aldehydes [48,49].

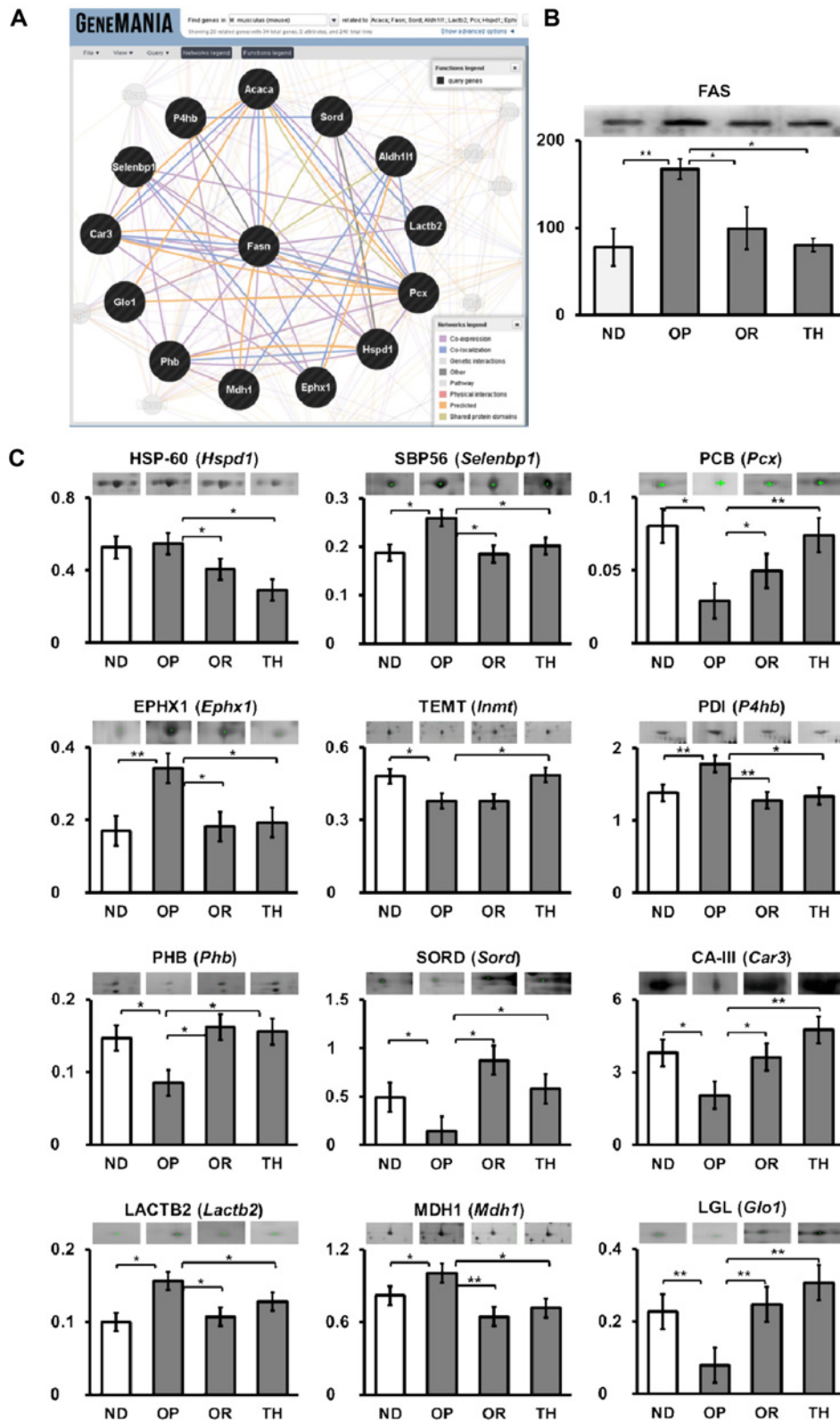


Fig. 8. Identification of differentially expressed liver proteins linked to *Fasn* (FAS) expression. (A) GeneMANIA result window depicts close-up of interaction with *Fasn* (FAS), (B) Western blot analysis of FAS, (C) differential expression of identified proteins from 2-DE. Band density of Western blot image was calculated using KODAK 1D Image Analysis software, and relative intensity (%) demonstrated that protein levels were normalized to that of β -actin. Data shown are representative of three independent experiments. Statistical significance was determined by One-way ANOVA, where p values are * $p < 0.05$ and ** $p < 0.01$ for significance.

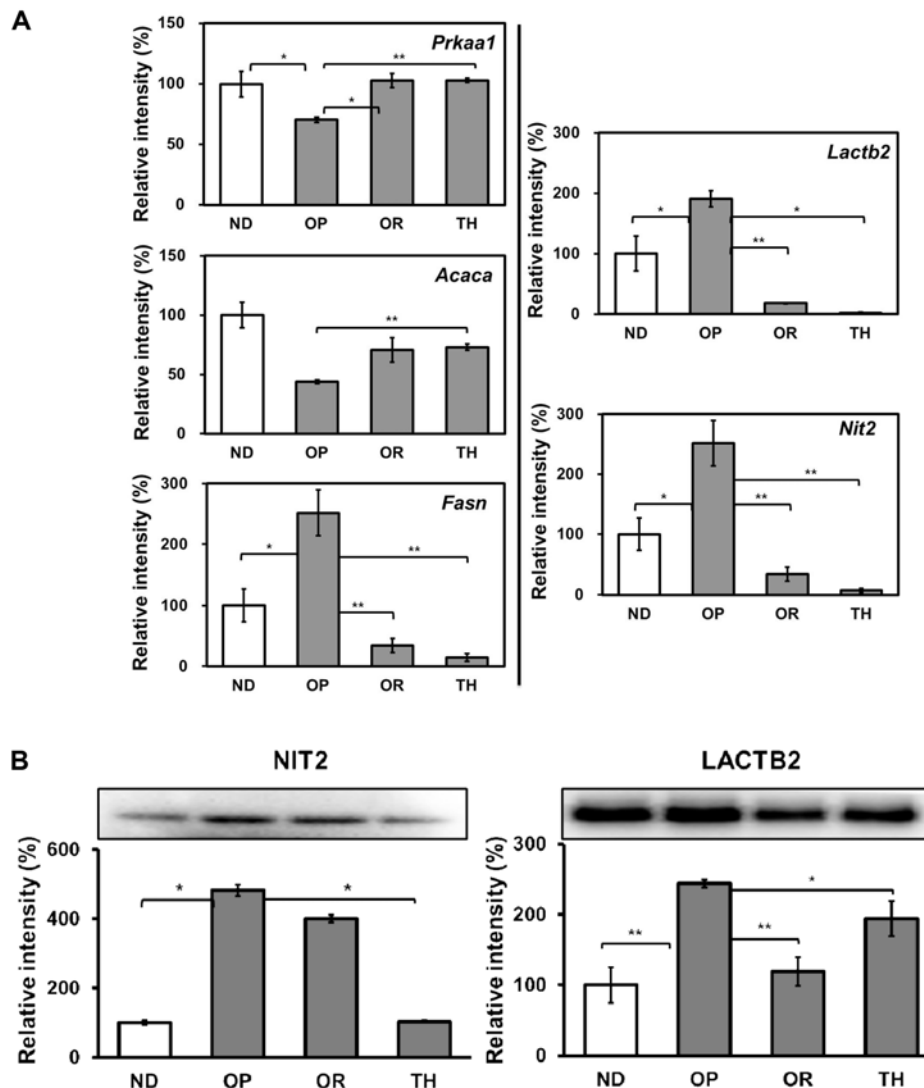


Fig. 9. Validation of proteomic data and predicted targets (*Lactb2*, *LACTB2* and *Nit2*, *NIT2*) generated by GeneMANIA using (A) real-time RT-PCR with three fatty acid metabolic genes and (B) Western blot analysis. Real-time PCR results are expressed as relative intensity normalized to *Gusb* gene expression and represent the mean $\pm$ S.E.M of triplicate independent experiments. Statistical significance was determined by One-way ANOVA, where * $p < 0.05$ and ** $p < 0.01$ for significance.

Therefore, reduced expression of α -AASA dehydrogenase in response to TH treatment may reflect inhibition of obesity caused by HFD consumption. Interestingly, three FDH protein spots showed increased levels in OP mice but reduced levels in both OR and TH mice, suggesting their roles in regulation of folate metabolism in the liver [50,51]. Further, epoxide hydrolase (EPHX1) showed increased expression in OP mice but reduced expression in both OR and TH mice, which supports previous findings that inhibition of EPHX1 may possibly attenuate HFD-induced hepatic steatosis *via* reduced systemic inflammatory status in mice [52]. Similarly, three protein spots of fructose-1,6-bisphosphatase (FBPase) showed increased expression in OP mice and reduced expression in TH mice. FBPase is a

regulatory enzyme in gluconeogenesis, and its levels are elevated upon obesity and dietary fat intake [53], suggesting its possible roles in other metabolic processes apart from glucose metabolism. This is the first study to identify FBPase as having differentially expressed forms in the livers of HFD-fed obese mice using 2-DE.

Interestingly, one protein spot of sorbitol dehydrogenase (SORD) showed up-regulated expression in OP mice, whereas another spot showed down-regulation. SORD participates in carbohydrate metabolism by converting sorbitol to fructose and thus protects diabetic tissues from glucose toxicity [54]. This discrepancy may be attributed to protein maturation and thereby possibly controls the rate of protein secretion in the liver. However, further studies are

required to directly test this hypothesis. To investigate potential connections between the identified proteins and three key fatty acid metabolic targets (AMPK, ACC and FAS), we performed network analysis using GeneMANIA (Figs. 6, 7, and 8). Down-regulation of AMPK (protein encoded by *Prkaa1*) as well as up-regulation of ACC (protein encoded by *Acaca*) and FAS (protein encoded by *Fasn*) was observed in the livers of OP mice. The lipid metabolism pathway is central to the integrated effects of AMPK in the liver and is closely associated with reduction of obesity [55]. Although AMPK activation has been shown to exert anti-obesity activity, its effect on metabolic processes upon an HFD treatment has not been fully understood. Thus, we first focused on AMPK partners of the identified proteins by proteomic analysis. In GeneMANIA analysis, *Prkaa1* interacted with eight targets in liver tissues of HFD-fed mice (Fig. 6). Moreover, of the eight predicted targets, six were classified as phosphoproteins by the DAVID bioinformatics tool.

Phosphorylation is one of the most studied PTM and is vital to cellular metabolic processing. Protein phosphorylation leads to reduction of pI and consequently can be observed in 2-DE gels. Among six phosphoprotein targets, HSP60 is well characterized in eukaryotic cells with regard to its function in protein folding [56]. Recent reports have described mitochondrial dysfunction in HFD-treated mice [57,58], and obesity is associated with mitochondrial dysfunction resulting from decreased levels of the mitochondrial chaperone HSP60 [59]. Furthermore, HSP60 plays important roles in both cell survival as well as regulation of mitochondrial function in obesity [59]. GeneMANIA analysis confirmed interactions of *Hspd1* with *Prkaa1*, as well as four other targets, including *P4hb* (PDI), *Anxa5* (ANXA5), *Nit2* (NIT2), and *Aldh1l1* (FDH), demonstrating a coordinated response during development of obesity caused by HFD treatment. Conclusively, AMPK activation in response to TH treatment may result in phosphorylation of predicted targets either directly or indirectly through unidentified substrates. This result suggests that the anti-obesity herbal medicine TH may contribute to phosphorylation of AMPK in HFD-induced obese mice.

GeneMANIA analysis also revealed that fatty acid metabolic protein acetyl-CoA carboxylase (ACC)-encoding gene *Acaca* interacts with eight targets in liver tissues of HFD-treated mice (Fig. 7). It has been well demonstrated that AMPK phosphorylates ACC to activate malonyl-CoA decarboxylase, thereby suppressing gene expression of lipogenic enzymes such as FAS in the liver [60]. Moreover, ACC inactivation (p-ACC) leads to increased fatty acid oxidation in the liver. Most genes in the network analysis (*Atp5h*, *Csnk2a1*, *Glo1*, *Krt18*, and *Pcx*)

belong to the acetylation class of DAVID functional categories, one of the most critical classes of PTM in protein function [61]. Earlier studies have demonstrated that acetylated proteins are associated with obesity, and their activity can be regulated for favorable cellular alteration of metabolic fuels [61,62].

Among the five interacting partners of *Acaca* belonging to the acetylation class of proteins, lactoylglutathione lyase (LGL encoded by *Glo1*), catalyzes the conversion of toxic α -oxo-aldehydes into corresponding α -hydroxy acids using L-glutathione (GSH) as a cofactor [63]. However, PTM and distinct functional assignment of this protein are largely unknown. A previous report demonstrated that PTM of LGL by GSH is important for many pathology such as tumor cell proliferation and diabetes and thus might need to be regulated in response to the redox state of tissues [64]. Inhibition of LGL by chemical reaction with its co-factor and the role of its intramolecular disulfides are expected to be important factors within the context of redox-dependent regulation of glucose metabolism in cells [64]. Moreover, acetylation of pyruvate carboxylase (PCB encoded by *Pcx*) has been observed in liver mitochondria from fasting *Sirt3* KO mice [65]. PCB catalyzes the initial reactions of glucose and lipid synthesis from pyruvate in the liver [66]. Our proteomic data indicate increased protein levels of both LGL and PCB in TH-treated mice, suggesting that TH may play a role in lipid homeostasis in liver tissues of HFD-induced obese mice. In addition, GeneMANIA analysis confirmed interactions among *Acaca* (ACC), *Pcx* (PCB), and *Glo1* (LGL) with two targets (*Car3/CA-III* and *Krt18/CK-18*), which suggests a coordinated response to HFD treatment.

Finally, GeneMANIA analysis predicted interactions between fatty acid synthase (FAS/*Fasn*) and 12 targets among HFD-treated mice (Fig. 8). Moreover, most genes belonged to acetylation class of DAVID functional categories. Among seven interacting partners of *Fasn* in acetylation class (*Glo1*, *Lactb2*, *Mdh1*, *Hspd1*, *Phb*, *Pcx*, and *Sord*), our proteomic data show that malate dehydrogenase (MDH1) was up-regulated in OP mice but down-regulated in OR/TH mice. MDH1 catalyzes reversible oxidation of malate to oxaloacetate and plays an important role in transporting NADH equivalents across the mitochondrial membrane [67]. A recent study demonstrated that acetylation of MDH1 is elevated in 3T3-L1 preadipocytes and promotes adipogenic differentiation [68], thereby providing evidence that adipogenic differentiation is regulated by acetylation of MDH1. MDH1 acetylation is a cross-talk mechanism between adipogenesis and the intracellular energy level [68]. Increased acetylation of HSP60 has been observed in *Sirt3* KO mitochondria, suggesting that changes in their acetylation status contribute to altered

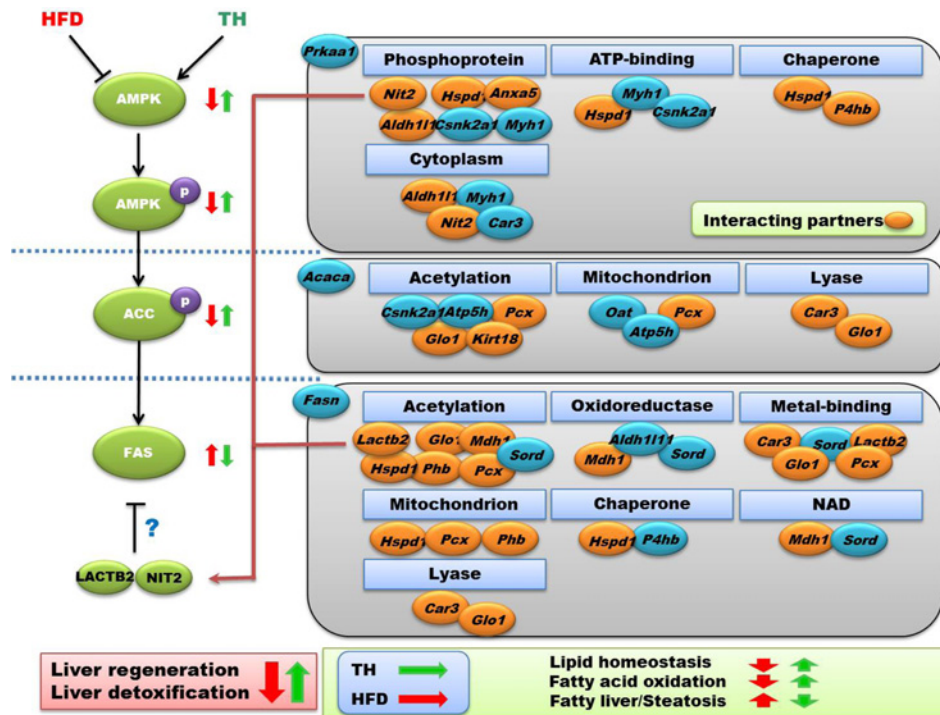


Fig. 10. Summary of differentially regulated proteins/genes in HFD-fed obese mice and response to anti-obesity herbal medicine (TH) treatment. For abbreviated protein names, see Abbreviations.

mitochondrial ROS production and response to oxidative stress [69]. Consequently, these results clearly suggest that the interacting partners of FAS/*Fasn* might be highly acetylated during fatty acid metabolism in HFD-induced obese mice.

GeneMANIA analysis further identified interactions among *Fasn* (FAS), *Mdh1* (MDH1), *Hspd1* (HSP-60), *Slenbp1* (SBP56), *Ephx1* (EPHX1), *Car3* (CA-III), *Pcx* (PCB), *Lactb2* (LACTB2), *Phb* (PHB), and *Glo1* (LGL) in HFD-induced OP mice. As another interacting partner of *Fasn*, *Slenbp1* was identified as a defense/immunity protein by the PANTHER classification tool. In our proteomic study, we observed up-regulation of selenium-binding protein (SBP) 1 but down-regulation of SBP2 in HFD-induced OP mice. Interestingly, these proteins showed opposite expression patterns between OP and OR/TH mice. The human SBP-encoding gene has been suggested to mediate intracellular transport of selenium [70,71]. Further, SBP1 has been proposed to serve as a marker in colonic cell differentiation [72] and plays a role in the proteasomal degradation pathway [73]. It was previously demonstrated that SBP2 has a “sensor” function in the nucleus where it triggers transcription of genes related to cell repair or promotion of necrosis/apoptosis [74]. A previous study demonstrated that both transcription and protein levels of SBP2 are decreased in the liver of HFD-induced mice [75]. Therefore, altered protein expression levels of SBP1 and SBP2 may

play important roles in lipid metabolism and can be used as potential targets for obesity development in HFD-induced obese mice.

One of the most remarkable outcomes of this study is the confirmation of a novel association of beta-lactamase-like protein 2 (LACTB2 encoded by *Lactb2*) with omega-amidase NIT2 (NIT2 encoded by *Nit2*) in fatty liver tissues of obese mice (Fig. 10). We detected up-regulation of both LACTB2 and NIT2 in HFD-induced obese mice, followed by their down-regulation upon TH treatment. NIT2 is a well-established tumor suppressor protein that potentially removes cellular toxic intermediates by converting α -ketoglutarate and α -ketosuccinamate into biologically useful α -ketoglutarate and oxaloacetate, respectively [76]. Although LACTB2, predicted as a zinc-binding hydrolase, has been identified in proteomic analysis of liver tissues from rats and humans [77,78], little is known about its role in obesity [79]. Therefore, our findings may provide clues to a possible link between LACTB2 and NIT2 in obesity development in mammals.

Among various ingredients of TH, *Schisandrae fructus*, *Herba ephedrae*, and *Radix platycodi* have reported to exhibit protective effects against hyperlipidemia and hepatosteatosis [80–84]. Moreover, *Radix platycodi* is also one of the key compounds of TH as it improves obesity and insulin resistance in obese mice *via* the activation of AMPK/ACC pathways and reduces of adipocyte

differentiation [84]. We also obtained the results that TH may modulate fatty acid metabolizing proteins/genes involved in obesity and hepatic injury, particularly *via* the AMPK pathway. In conclusion, the current combined proteomic and network analysis data confirm that the anti-obesity herbal medicine TH exhibits anti-obesity effects by modulating fatty acid metabolizing proteins/genes, particularly *via* the AMPK pathway (Fig. 10). Most targeted proteins/genes in this study modulated to improve lipid metabolism in response to TH treatment. Current approaches provide an understanding of the complex cellular networks and regulation of pathways, as well as, identifying novel proteins/genes involved in potential therapeutic strategies for obesity treatment.

Acknowledgements

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Nomenclature

2-DE	: Two-dimensional electrophoresis
α -AASA dehydrogenase/ <i>Aldh7a1</i>	: Alpha-amino adipic semialdehyde dehydrogenase
ACC/ <i>Acaca</i>	: Acetyl-CoA carboxylase
ANXA5/ <i>Anxa5</i>	: Annexin A5
AMPK/ <i>Prkaa1</i>	: AMP-activated protein kinase
ATP5H/ <i>Atp5h</i>	: ATP synthase subunit D
CA-III/ <i>Car3</i>	: Carbonic anhydrase 3
CK-18/ <i>Krt18</i>	: Endo B cytokeratin
CK II α / <i>Csnk2a</i>	: Casein kinase II subunit alpha
EPHX1/ <i>Ephx1</i>	: Epoxide hydrolase
FAS/ <i>Fasn</i>	: Fatty acid synthase
FDH/ <i>Aldh1l1</i>	: Cytosolic 10-formyltetrahydrofolate dehydrogenase
<i>Gusb</i>	: Glucuronidase beta
HFD	: High fat diet
HSP60/ <i>Hspd1</i>	: Heat shock protein 60
LACTB2/ <i>Lactb2</i>	: Beta-Lactamase-like protein 2
LGL/ <i>Glo1</i>	: Lactylglutathione lyase
MDH1/ <i>Mdh1</i>	: Malate dehydrogenase
MYH1/ <i>Myh1</i>	: Myosin 1
NIT2/ <i>Nit2</i>	: Omega-amidase NIT2
OAT/ <i>Oat</i>	: Ornithine aminotransferase
OP	: Obesity-prone
OR	: Obesity-resistant

PCB/ <i>Pcx</i>	: Pyruvate carboxylase
PDI/ <i>P4hb</i>	: Protein disulfide-isomerase precursor
PHB/ <i>Phb</i>	: Prohibitin
PTM	: Post-translational modification
SBP/ <i>Selenbp</i>	: Selenium-binding protein
SORD/ <i>Sord</i>	: Sorbitol dehydrogenase
TCP-1 β / <i>Cct2</i>	: T-Complex protein 1 subunit beta
TEMT/ <i>Inmt</i>	: Indolethylamine <i>N</i> -methyltransferase

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